

Metric		Definition	Instantiation (testbed)	Axis stressed
Advantage-of-prediction curve		closed-loop success <i>gain</i> of a predictive policy over a matched direct-VLA baseline, reported per capability as a curve, not a single endpoint	a capability-sliced protocol over LIBERO [3] and CALVIN [4] that runs a world-model policy and a VLA policy under one harness	γ family, δ metric
Counterfactual accuracy	accu-	task success on episodes that require inference over unobserved or hypothetical dynamics (occlusion, object permanence, “what if”)	WorldPrediction [1] plus occlusion / memory splits in the style of LIBERO-Mem [2]	β capability
Prediction-to-action fidelity gap	fi-	the difference between a world model’s open-loop generation quality and its closed-loop task success (“visual quality is not task success”)	bridge protocols such as WorldSimBench [5] and World-in-World [6]	α mode
Per-capability contrast coverage		the fraction of capabilities for which a benchmark actually runs a VLA-vs-world-model head-to-head under a single protocol	audits of the landscape (Table ??) against a declared contrast schema	γ family, ε catalog

Table 1. **An actionable protocol for the future agenda:** four measurable quantities, their definitions, and concrete testbeds. The first anchors the advantage-of-prediction question (δ) on LIBERO / CALVIN-style suites; the second targets the near-unmeasured counterfactual capability; the third quantifies the prediction-to-action gap the bridges expose; the fourth audits per-capability contrast coverage across the landscape. Each is reported as a distribution or curve, never a single headline number.